SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO2: Apply data pre-processing and visualization techniques using Python libraries for effective data handling (BL4)

QUESTIONS: 20 **Total Marks:100**

Annexure A

MCQ

Code of Conduct:

I SIVA SUBRAMANIYAM B/192572089 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)	Level	Marks
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept tested in the MCQ	Shows good understanding with minor conceptual gaps	Shows basic understanding with some misconceptions	Shows poor or incorrect understanding of the concept		
Correct Option Selection	**30**	Selects the correct answer accurately without confusion	Selects the correct answer with slight uncertainty	Selects a partially relevant or close option but not fully correct	Selects an incorrect or irrelevant option		
Analytical Reasoning	**20**	Clearly analyzes the question and identifies the logic behind the correct option	Provides reasonable analysis with minor errors	Shows limited reasoning or incomplete analysis	Provides no valid reasoning or guesses the answer		
Elimination of Wrong Options	**15**	Correctly identifies why all incorrect options are wrong	Identifies most incorrect options with minor explanation gaps	Identifies only some wrong options with weak justification	Does not identify or explain incorrect options		
Clarity and Response Presentation	**10**	Response is clear, concise, and well-organized	Response is understandable with minor clarity issues	Response is somewhat unclear or poorly structured	Response is confusing, incomplete, or missing		

Annexure A

MCQ

A CSV file is imported into Pandas, but the first row of actual data is wrongly treated as column names. What is the best correction?

A. Use `header=None` while reading the file
B. Use `dropna()` after reading the file
C. Use `sort_values()` before reading the file
D. Use `groupby()` after reading the file

Answer: A

2. Selecting Required Columns

A dataset has 25 columns, but only `Age`, `Salary`, and `Department` are required for analysis. Why should only these columns be selected?

A. To increase duplicate values
B. To reduce unnecessary data processing
C. To remove all missing values automatically
D. To convert categorical data into numerical data

Answer: B

3. Conditional Row Selection

A manager wants employees from the IT department with salary above ₹60,000. Which command is correct?

A. `df[(df['Department']=='IT') & (df['Salary'] > 60000)]`
B. `df[df['Department']=='IT' & df['Salary'] > 60000]`
C. `df[(df['Department']=='IT') or (df['Salary'] > 60000)]`
D. `df['Department'=='IT' and 'Salary'>60000]`

Answer: A

A student dataset is filtered using:

```
df[df['Marks'] >= 50]
```

What does this operation return?

A. Students who failed
B. Students who scored exactly 50 only
C. Students who scored 50 or above
D. Students whose marks are missing

Answer: C

A dataset has 40% missing values in one column. What is the most analytical concern before filling missing values?

A. Whether the column name is short
B. Whether missingness may affect the reliability of analysis
C. Whether the column can be sorted alphabetically
D. Whether the column has duplicate values only

Answer: B

Mean imputation is most suitable when:

A. The data contains extreme outliers
B. The numerical data is approximately symmetric
C. The column contains only text values
D. More than 90% of values are missing

Answer: B

A salary column contains a few very high executive salaries. Some salary values are missing. Which imputation method is more suitable?

A. Mean
B. Median
C. Mode
D. Drop the entire DataFrame

Answer: B

A City column contains missing values. Which method is usually suitable?

A. Replace missing values with the mean city
B. Replace missing values with the most frequent city
C. Sort cities in descending order
D. Convert all cities into numbers directly

Answer: B

9. Dropping Missing Values

A dataset has 10,000 rows, and only 12 rows contain missing values. What is a reasonable approach?

A. Delete all columns
B. Drop the 12 incomplete rows
C. Replace all values with zero
D. Ignore the missing values permanently

Answer: B

A company wants to identify the top 5 highest-paid employees. Which sequence is most appropriate?

A. Group by department and calculate mean
B. Sort salary in descending order and select first 5 rows
C. Fill missing values with zero and stop analysis
D. Sort employee names alphabetically

Answer: B

A school wants to sort students first by Class and then by Marks in descending order within each class. Which command is correct?

A. df.sort_values(['Class', 'Marks'], ascending=[True, False])
B. df.sort_values(['Class', 'Marks'], ascending=False)
C. df.groupby(['Class', 'Marks'])
D. df.rank(['Class', 'Marks'])

Answer: A

A sales dataset contains Region, Product, and Sales. To find total sales for each region and product combination, which operation is best?

A. df.groupby(['Region', 'Product'])['Sales'].sum()
B. df.sort_values(['Region', 'Product'])
C. df['Sales'].rank()
D. df.dropna(['Region', 'Product'])

Answer: A

A college wants to compare average marks department-wise. Which result best supports this analysis?

A. Total number of rows in the dataset
B. Mean marks grouped by department
C. Alphabetical order of student names
D. Number of missing values in student names

Answer: B

A company wants department-wise average salary and maximum salary. Which command is suitable?

A. df.groupby('Department')['Salary'].agg(['mean', 'max'])
B. df.sort_values('Department')['Salary']
C. df['Salary'].fillna(['mean', 'max'])
D. df.rank('Department')['Salary']

Answer: A

Two students have the same mark. What does ranking help identify?

A. It removes one student automatically
B. It assigns relative positions while handling ties based on rank method
C. It replaces both marks with missing values
D. It sorts only text columns

Answer: B

A company wants to rank employees by salary within each department. Which command is correct?

A. df['Rank'] = df.groupby('Department')['Salary'].rank(ascending=False)
B. df['Rank'] = df['Department'].rank('Salary')
C. df['Rank'] = df.sort_values('Salary')
D. df['Rank'] = df.groupby('Salary')['Department']

Answer: A

A researcher wants to compare average marks of five departments. Which visualization is most suitable?

A. Bar chart
B. Scatter plot
C. Histogram
D. Line chart

Answer: A

A company records monthly profit for 24 months. Which chart gives the best analytical understanding of growth or decline?

A. Line chart
B. Pie chart
C. Box plot
D. Histogram

Answer: A

A dataset contains Advertising Budget and Sales Revenue. Which visualization is most suitable to analyze whether higher advertisement spending increases sales?

A. Scatter plot
B. Pie chart
C. Histogram
D. Stacked bar chart

Answer: A

A dataset contains employee salaries, and the analyst suspects some extremely high salaries. Which visualization is most useful?

A. Box plot
B. Pie chart
C. Line chart
D. Count plot

Answer: A